Network Reliability investigation

HackerRank

An ML platform has `n` services connected by one-way network links. Each link has a positive latency. A diagnostic request must travel from service `1` to service `n`.

Among all routes with the minimum total latency, determine:

1. The minimum latency.
2. The number of minimum-latency routes, modulo `1,000,000,007`.
3. The minimum number of links used by any minimum-latency route.
4. The maximum number of links used by any minimum-latency route.

Two routes are different if their sequences of links differ. The network may contain cycles, self-links, and duplicate links.

If service `n` cannot be reached from service `1`, report `-1 0 0 0`.

Examples

Example 1

Input:

```
4 5
1 2 1
1 3 1
2 4 1
3 4 1
1 4 2
```

Output:

```
2 3 1 2
```

Explanation: The minimum latency from service 1 to service 4 is 2. Three routes achieve it: `1 -> 2 -> 4`, `1 -> 3 -> 4`, and the direct link `1 -> 4`. The shortest of them uses 1 link and the longest uses 2 links.

Example 2

Input:

```
3 1
1 2 5
```

Output:

```
-1 0 0 0
```

Explanation: No chain of links ever reaches service 3, so the request cannot be delivered.

Example 3

Input:

```
3 4
1 2 1
1 2 1
2 3 2
3 3 1
```

Output:

```
3 2 2 2
```

Explanation: The two duplicate links `1 -> 2` give two different routes, both with latency 1 + 2 = 3 and 2 links. The self-link on service 3 can never be part of a minimum-latency route, because every latency is positive.

Input Format

The first line contains two integers `n` and `m` — the number of services and the number of directed links.

The next `m` lines each contain three integers `a`, `b`, and `latency` — a one-way link from service `a` to service `b` with the given latency.

Services are numbered from `1` to `n`.

Constraints

- `1 ≤ n ≤ 100,000`
- `0 ≤ m ≤ 200,000`
- `1 ≤ latency ≤ 1,000,000,000`
- `1 ≤ a, b ≤ n`
- Link latencies are positive integers.
- Self-links and duplicate links may occur.

Output Format

If service `n` is reachable from service `1`, print four space-separated integers: the minimum latency, the number of minimum-latency routes modulo `1,000,000,007`, the minimum number of links among those routes, and the maximum number of links among those routes.

If service `n` is unreachable, print `-1 0 0 0`.

If `n = 1`, the request is already at its destination: the route uses zero links, so print `0 1 0 0`.

Sample Input 0

```
4 5
1 2 1
1 3 1
2 4 1
3 4 1
1 4 2
```

Sample Output 0

```
2 3 1 2
```